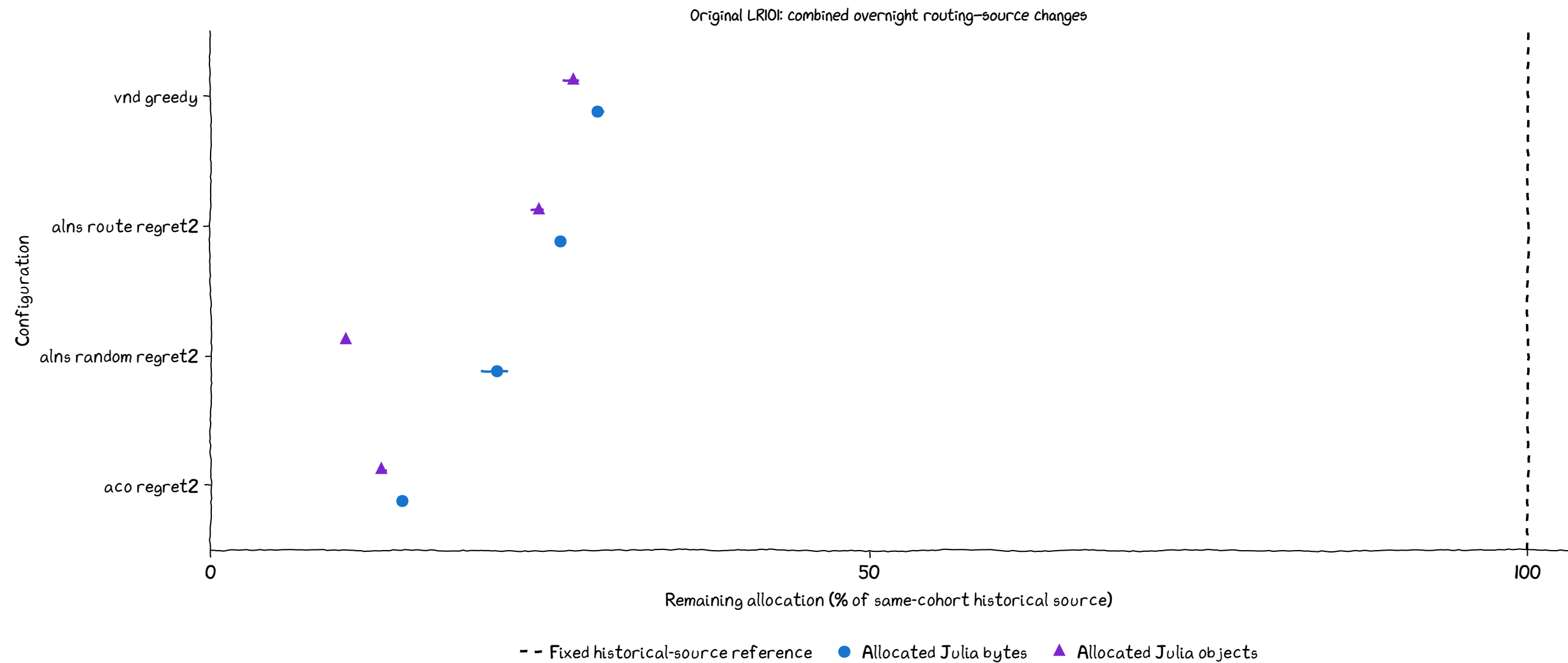




12 paired paths: 4 configurations × 3 seeds; 4,096 steps per lane × 2 workers. Historical routing source from the start of the night; all other current application and package sources shared. Markers: mean paired ratio; bars: full seed range. Complete substantive trace/routes/RNG work and original validators match; zero compilation/recompilation. Frozen V7 integration remains provisional for a separate generic QUBO constructor regression. Concurrent timing; no controlled speed or quality claim.